Physics 477, Homework #1

Due 2026 Jan 16

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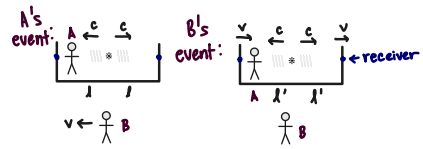
Date: 1/25/26

Class: MWF 11a

1. Morin 1.1: Consistency with Lv/c^2

   Show that the $t_R - t_L$ difference of the times in Eq. (1.6) (where l' is half the length of the train in the ground frame) is consistent with the Lv/c^2 rear-clock-ahead result (where L is the proper length of the train).

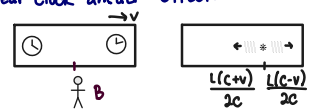


time to hit receiver from B's pov

↳ $t_L = \frac{l'}{c+v}$, $t_R = \frac{l'}{c-v}$: $t_L < t_R$

⇒ not simultaneous!

"rear clock-ahead" effect:



extra distance traveled by $\mathcal{O}_L$: $\frac{L(c+v)}{2c} - \frac{L(c-v)}{2c} = \frac{Lc+Lv-[Lc-Lv]}{2c}$

↳ $\frac{2Lv}{2c} = \frac{Lv}{c}$

time $= \frac{\Delta x}{v}$

so, speed ↳ $\frac{Lv}{c^2}$

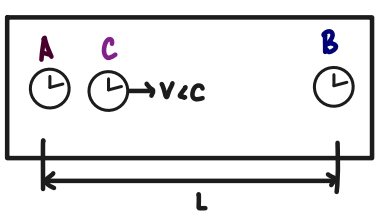
$t_R - t_L$: $\frac{l'}{c-v} - \frac{l'}{c+v} \Rightarrow \frac{l'}{c-v}\frac{(c+v)}{(c+v)} - \frac{l'}{c+v}\frac{(c-v)}{(c-v)}$

↳ $\frac{l'(c+v)}{c-v(c+v)} - \frac{l'(c-v)}{c+v(c-v)} \Rightarrow \frac{\cancel{cl'}+vl'}{c^2+\cancel{vc}-\cancel{vc}-v^2} - \frac{\cancel{cl'}-vl'}{c^2-\cancel{vc}+\cancel{vc}-v^2}$

$\Rightarrow \frac{2vl'}{c^2-v^2} \Rightarrow \frac{v(2l')}{c^2-v^2}$: $2l' = L$: $\frac{Lv}{c^2-v^2} \simeq \gamma\frac{Lv}{c^2}$

2. Morin 1.3: Synchronizing Clocks

   Two synchronized clocks, A and B, are at rest in a given frame, a distance L apart. A third clock, C, is initially located right next to A. All three clocks have initial readings of zero, and then C is moved very slowly (with speed $v \ll c$) from A to B. Show that its final reading can be made arbitrarily close to B's, by making v be sufficiently small. (The Taylor series $\sqrt{1-\epsilon} \approx 1 - \epsilon/2$ will come in handy.)



$t_C = \frac{t_B}{\gamma}$ (time dilation)

where $\gamma = \frac{1}{\sqrt{1-v^2/c^2}}$

so it follows: $t_C = t_B\sqrt{1-v^2/c^2}$

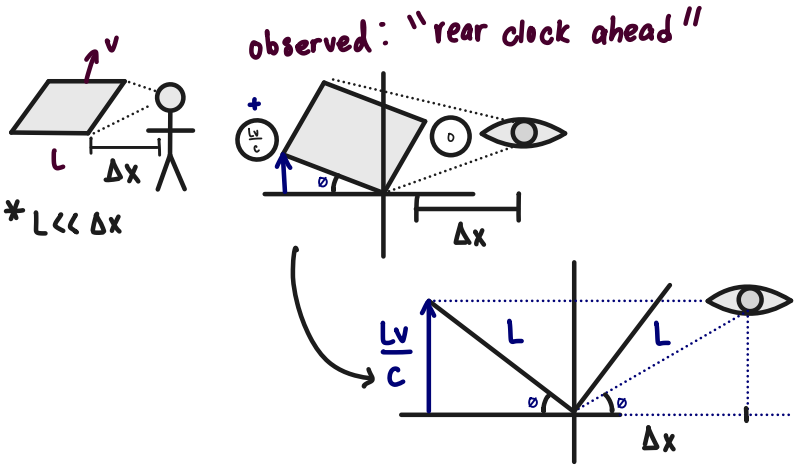
with $v^2/c^2 = \epsilon$ in the taylor series we see: $\gamma \simeq 1 - \frac{v^2}{2c^2}$

or that: $t_C = t_B\left(1 - \frac{v^2}{2c^2}\right)$ such that if $v <<< c$

then $t_C - \lim_{v\to 0} t_B\left(1 - \frac{v^2}{2c^2}\right) = 0$ or $t_C = t_B$

3. Morin 1.4: Rotated square

A square with sides of proper length L flies past you at speed v, in a direction parallel to two of its sides. You stand in the plane of the square. When you see the square at its nearest point to you (see Fig. 1.47), show that it *looks* to you like it is rotated instead of contracted, and find the apparent angle of rotation. Assume that L : is small compared with the distance between you and the square. (This setup is one of the few cases where we are actually concerned with the time it takes light to travel to your eye.)

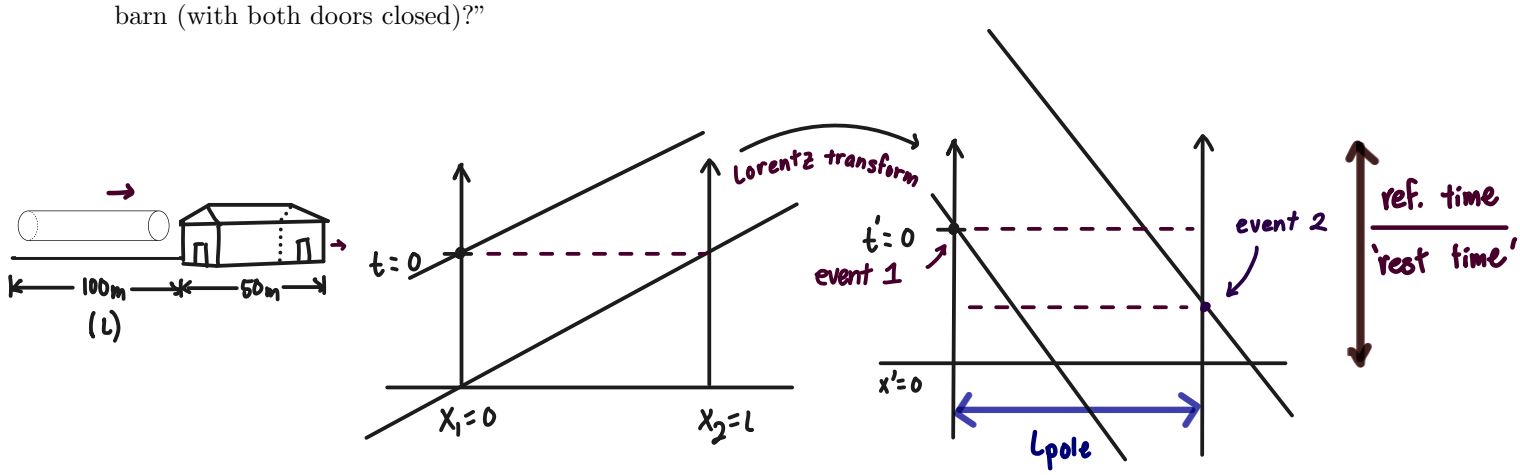


$$*\Delta x = v \cdot t = v\left(\frac{L}{c}\right) = \frac{Lv}{c}$$

$$\phi = \sin^{-1}\left(\frac{\left[\frac{Lv}{c}\right]}{L}\right) = \sin^{-1}\left(\frac{v}{c}\right)$$

4. Morin 1.6: Pole in barn

A pole with proper length L moves rightward with speed v through a barn, also with proper length L. Assume that initially the door at the left end of the bar is open and the door at the right end is closed. Just after the left end of the pole enters the barn, the left door closes. And just before the right end of the pole leaves the barn, the right door opens. How would you respond to the following question: "Is the pole ever completely inside the barn (with both doors closed)?"

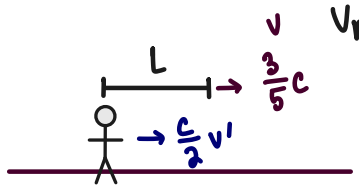


from the reference frame of an observer in the barn, the pole undergoes length contraction and fits and both appear shut.

from the reference frame of an observer on the pole, the simultaneous shutting of the doors does <u>not</u> occur due to the effect of time dilation. The 1st door shuts only <u>after</u> the 2nd door opens and the pole never is fully inside w/ both doors shut.

5. Morin 1.12: Transforming the length

A stick moves rightward with speed $\frac{3}{5}c$ with respect to the ground. The length of the stick *in the ground frame* is L. You move rightward with speed $c/2$ with respect to the ground. What is the length of the stick in your frame?



$$V_{rel.} = \frac{v - v}{1 - \frac{vv}{c}} \Rightarrow \frac{\frac{1}{10}c}{1 - 3/10} \Rightarrow \frac{\frac{1}{10}c}{\frac{7}{10}} = \frac{1}{7}c$$

$$\hookrightarrow L = L_0\sqrt{1 - \frac{V_{rel}^2}{c^2}} \Rightarrow L = L\sqrt{1 - \frac{(\frac{c}{7})^2}{c^2}} = L\sqrt{1 - \frac{1}{49}} = L\sqrt{\frac{48}{49}}$$

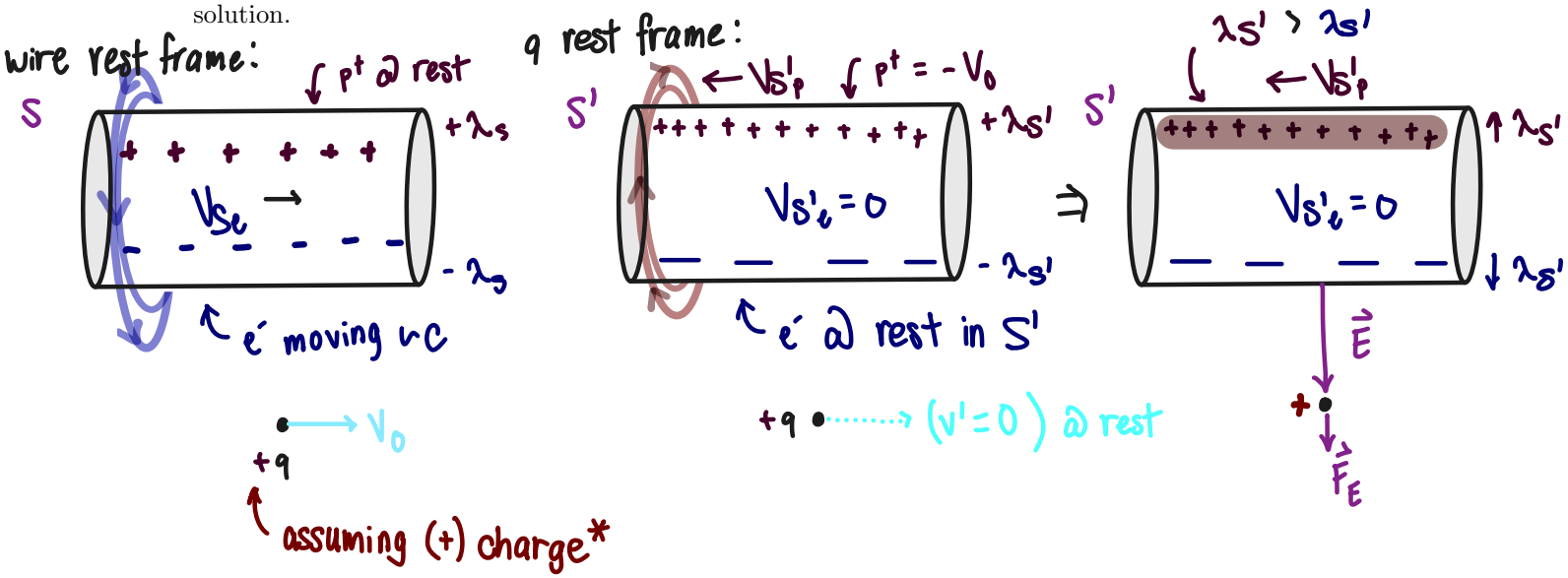
6. Morin 1.13: Magnetic Force

This problem demonstrates how the magnetic force arises from the combination of the electric force and length contraction. The interpretation of this problem (discussed in the solution) requires a familiarity with basic concepts of electricity and magnetism, although the problem itself does not.

Consider a current-carrying straight wire. the wire is neutral, that is, it has the same number of negatively charged electrons and positively charged protons in any given volume, on average. (If it weren't neutral, it would attract or repel electrons, thereby producing neutrality.) In any current-carrying wire, the current is caused by electrons moving along the wire; the protons are bolted down. So we have the situation shown in Fig. 1.50. (Ignore the charge q for a moment.) In reality, the electrons and protons are distributed throughout the wire, but we have drawn them separated for clarity. The qualy and opposite charge densities $\pm\lambda_0$ (charge per meter) are indicated. The electron's speed is v_0.

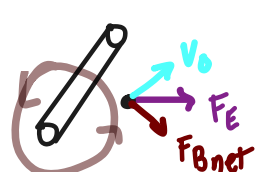
Now consider an electric charge q near the wire. If the charge q is at rest, then since the wire is neutral, q will feel no force. But let's assume that the charge q is moving to the right with speed v, as shown in Fig. 1.50. What are the charge densities of the protons and electrons in q's rest frame? What then is the net charge density of the wire in q's frame?

You will find that the charge density is nonzero, which means that the wire exerts a force on q in q's frame (and hence also in the original frame). Returning to the original frame, we conclude that an electric charge q that is *at rest* near a neutral current-carrying wire feels no force, whereas a charge q that is *moving* near a neutral current-carrying wire *does* feel a force. This force is known as the *magnetic force*. We will discuss it in the solution.



$\vec{F}_B = qv \times \vec{B}$ ⊗ into page < $\vec{F}_B = qv \times \vec{B}$ ● out of page

$$\vec{F}_{net} = \overbrace{\left(\vec{F}_{B_S} + \vec{F}_{B_{S'}}\right)}^{\vec{F}_{B_{net}}} + \vec{F}_E \Rightarrow$$



$$\hookrightarrow \lambda_{S'p^+} = \gamma_S \lambda_{p^+}$$

$$u = \frac{V_{se} - V_{s'e}}{1 - [V_{se}(V_{s'e})/c^2]}$$

7. Morin 1.14: Pythagorean triples

Let (a, b, h) be a Pythagorean triple. (We'll use h to denote the hypotenuse, instead of c, for obvious reasons.) Consider the relativistic addition or subtraction of two speeds with β values of $\beta_1 = a/h$ and $\beta_2 = b/h$. ($\beta \equiv v/c$ is a speed's fraction of the speed of light.) Show that the numerator and denominator of the result are the leg and hypotenuse of another Pythagorean triple, and find the other leg. What is the associated γ factor?

speed of moving frame S'

$$u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$$

speed of object measured in moving frame S'

speed of object measured in rest frame S

$$\beta_1 = \frac{a}{h} \qquad \beta_2 = \frac{b}{h} \qquad \beta = \frac{v}{c}\,* \qquad a^2 + b^2 = h^2$$

$$\beta = \frac{\frac{a}{h} + \frac{b}{h}}{1 + \frac{\left(\frac{a}{h}\cdot\frac{b}{h}\right)}{h^2}} \Rightarrow \frac{h(a+b)}{h^2 + ab} \Rightarrow (h(a+b))^2 + x^2 = (h^2 + ab)^2$$

$$\hookrightarrow \underbrace{(h^2a^2 + h^2ab + b^2h^2)}_{a'^2} + \underbrace{x^2}_{b'^2} = \underbrace{h^4 + 2h^2(ab) + a^2b^2}_{c'^2}$$

$$h^2(a^2 + ab + b^2) + x^2 = h^4 + 2h^2ab + a^2b^2$$

$$a^2 + b^2 = h^2$$

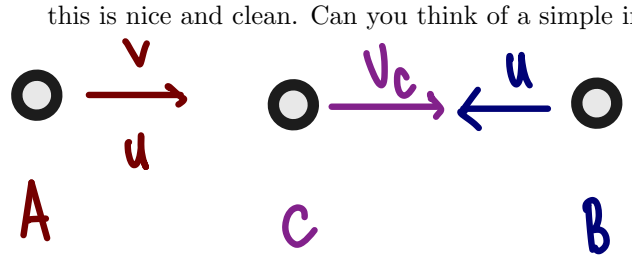
$$\hookrightarrow h^2(h^2 + ab) + x^2 = h^4 + 2h^2ab + a^2b^2 \Rightarrow x = ab!$$

$$*\,\gamma = \frac{hyp}{adj} \quad \text{for triples!}$$

$$\gamma = \frac{h^2 + ab}{h(a+b)}$$

8. Morin 1.17: More equal speeds

A travels at speed v with respect to the ground, and B is at rest, as shown in Fig. 1.52. How fast should C travel so that she sees A and B approaching her at the same speed? In the ground frame (B's frame), what is the ratio of the distances CB and AC? (Assume that A and C arrive at B at the same time.) The answer to this is nice and clean. Can you think of a simple intuitive explanation for the result?



speed of moving frame S'

$$u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$$

speed of object measured in moving frame S'

speed of object measured in rest frame S

need to find v when C is @ S

$\rightarrow v_c = u$

$u = v_c$ wrt B

quadratic!

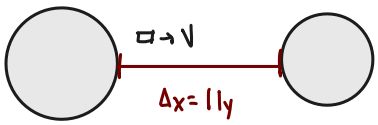
$$v = \frac{v_c + u}{1 + \frac{v_c u}{c^2}} = \frac{2v_c}{1 + \frac{v_c^2}{c^2}} \cdot \left(1 + \frac{v_c^2}{c^2}\right) = v\left(1 + \frac{v_c^2}{c^2}\right) \Rightarrow v\left(1 + \frac{v_c^2}{c^2}\right) = 2v_c \Rightarrow v + v\frac{v_c^2}{c^2} - 2v_c = 0$$

$$\hookrightarrow v\frac{v_c^2}{c^2} - 2v_c + v = 0 : v_c = \frac{2 \pm \sqrt{4 - 4\left(\frac{v}{c^2}\right)(v)}}{\frac{2v}{c^2}} \Rightarrow \frac{2 \pm \sqrt{4\left(1 - \left(\frac{v}{c^2}\right)v\right)}}{\frac{2v}{c^2}} \Rightarrow \frac{2 \pm 2\sqrt{1 - \frac{v^2}{c^2}}}{\frac{2v}{c^2}} \Rightarrow \frac{1 \pm \sqrt{1 - v^2/c^2}}{\frac{v}{c^2}}$$

$$*\,\sqrt{1 - 1/\gamma} = \sqrt{1 - v^2/c^2}, \text{ so: } v_c = \frac{1 \pm \sqrt{1 - 1/\gamma}}{v/c^2} \Rightarrow = 1 \pm \sqrt{1 - 1/\gamma} \Rightarrow v_c = \frac{c^2}{v}\left(1 \pm \sqrt{1 - 1/\gamma}\right)$$

9. Morin 1.22: Effectively speed c

A rocket flies between two planets that are one light-year apart. What should the rocket's speed be so that the time elapsed on the captain's watch is one year?



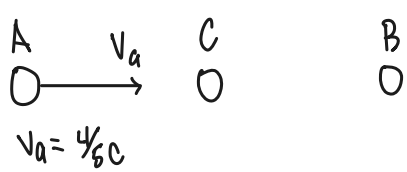
$\Delta t = \gamma t_0$

find V s.t. in $t = 1y$, $\Delta x = 1 ly$:

$V = \frac{\Delta x}{t_0} \Rightarrow \gamma V = \frac{\Delta x}{t_0}$: $\gamma = \frac{V}{\sqrt{1 - V^2/c^2}} = \frac{\Delta x}{t_0}$: $\frac{1 ly}{1y} = c$

$\frac{V}{\sqrt{1 - V^2/c^2}} = c$, thus: $V = \frac{c}{\sqrt{1 - V^2/c^2}}$, $V^2 = c^2\left(1 - \frac{V^2}{c^2}\right) = c^2 - V^2$

$\hookrightarrow 2V^2 = c^2 \Rightarrow V^2 = \frac{c^2}{2} \Rightarrow V = \frac{c}{\sqrt{2}}$

10. Morin 1.42: Equal speeds

A travels at speed $\frac{4}{5}c$ toward B, who is at rest. C is between A and B. How fast should C travel so that she sees both A and B approaching her at the same speed? (This problem is a special case of Problem 1.17. solve this one from scratch, but feel free to check your answer with the one from Problem 1.17).